

Structured Training Program: Wi-Fi

Module 4: WLAN MAC Layer

1. What is the significance of MAC layer and in which position it is placed in the OSI model?

The MAC (Medium Access Control) layer is a sublayer of the Data Link Layer (Layer 2) in the OSI model. In the IEEE 802 architecture (used by both Ethernet 802.3 and WiFi 802.11), the Data Link Layer is divided into two sublayers:

1. Upper sublayer: Logical Link Control (LLC) — provides a common interface to upper layers (e.g., Network Layer) and handles flow control, error control, and multiplexing.
2. Lower sublayer: MAC — specific to the medium and responsible for managing access to the shared physical medium.

Position in OSI Model:

OSI Layer 2: Data Link Layer - MAC sublayer (802.11 defines this) + LLC sublayer.

Significance of the MAC Layer in WiFi (802.11):

- The wireless medium is shared and half-duplex (devices cannot reliably detect collisions while transmitting, unlike wired Ethernet's CSMA/CD). MAC solves this using CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).
- It coordinates fair access among multiple devices, handles addressing in a wireless environment (up to 4 addresses per frame due to AP involvement), manages frame formatting, error detection (via FCS), retransmissions, fragmentation, and security-related coordination.
- It enables key WiFi features: roaming, power saving, authentication/association, QoS (via EDCA), and reliable data delivery over an unreliable radio link.
- Without a robust MAC, devices would collide frequently, leading to poor performance in dense or hidden-node scenarios.

2. Describe the frame format of the 802.11 MAC header and explain the purpose of each field

The 802.11 MAC frame (MPDU — MAC Protocol Data Unit) consists of:

- MAC Header (variable, typically 24–36 bytes)
- Frame Body (0–2312 bytes, carries higher-layer data or management info)
- FCS (Frame Check Sequence) — 4 bytes (CRC-32 for error detection)

General 802.11 MAC Frame Structure

Field	Length	Purpose
Frame Control	2 bytes	Controls frame type, direction, fragmentation, power management, security, and more (detailed below).
Duration/ID	2 bytes	Indicates NAV (Network Allocation Vector) duration in μ s for virtual carrier sensing, or Association ID in certain control frames. Helps with collision avoidance.
Address 1	6 bytes	Receiver Address (RA) — immediate recipient of the frame (always present).
Address 2	6 bytes	Transmitter Address (TA) / Source Address — sender of the frame (always present).
Address 3	6 bytes	Varies by To DS / From DS bits: usually Destination Address (DA) or BSSID (AP MAC).
Sequence Control	2 bytes	Fragment Number (4 bits) + Sequence Number (12 bits). Supports fragmentation and duplicate detection.

QoS Control	2 bytes	(Data frames only) — TID (Traffic Identifier) for QoS prioritization, ACK policy, etc.
HT/VHT/HE Control	0–4 bytes	(High Throughput and later) — Additional control for MIMO, beamforming, etc.
Frame Body	Variable	Payload: MSDU (data), management information, or null.
FCS	4 bytes	CRC-32 checksum for integrity check.

The combination of Type + Subtype determines the exact frame function. Address fields are interpreted differently based on To DS / From DS bits (up to 4 addresses in wireless bridging scenarios).

3. Please list all the MAC layer functionalities in all Management, Control and Data plane

The 802.11 MAC operates across three planes:

- i) Management Plane (Management Frames — Type 00)
 - Network Discovery & Maintenance: Beacon transmission/reception, Probe Request/Response, Scanning.
 - Joining the Network: Authentication (Open/Shared Key), Association/Reassociation, Disassociation/Deauthentication.
 - Roaming Support: Reassociation when moving between APs.
 - Synchronization: Timing Synchronization Function (TSF) via beacons.
 - Power Management Coordination: TIM (Traffic Indication Map), ATIM (Ad-hoc).
 - Security Setup: RSN Information Element exchange.
 - Spectrum Management: Channel switching, TPC (Transmit Power Control), DFS (Dynamic Frequency Selection).

- ii) Control Plane (Control Frames — Type 01)
 - Medium Access Coordination: RTS (Request to Send), CTS (Clear to Send) — solves hidden node problem.
 - Acknowledgments: ACK, Block ACK — confirms successful reception and triggers retransmissions.
 - Contention-Free Period Control: CF-End, CF-Poll (legacy PCF).
 - Protection & NAV Setting: Duration field usage for virtual carrier sensing.

- iii) Data Plane (Data Frames — Type 10)
 - Data Delivery: Carrying MSDU (higher-layer payload) as MPDU.
 - Fragmentation & Reassembly: Breaking large frames and reassembling.
 - QoS Support: Prioritization using EDCA (Enhanced Distributed Channel Access) with different ACs (Access Categories).
 - Aggregation: A-MSDU and A-MPDU for efficiency.
 - Retransmission & Error Recovery: Automatic retransmission on missing ACK.
 - Power-Save Delivery: Buffered data delivery using PS-Poll or U-APSD.

Overall MAC Responsibilities:

1. CSMA/CA channel access (DCF/HCF).
2. Addressing and filtering.
3. Frame formatting and validation (FCS).
4. Security (encryption, integrity).
5. Rate adaptation coordination.

4. Explain the scanning process and its types in detail

Scanning is the process by which a client station (STA) discovers available WiFi networks (BSSs) before joining one.

There are two types:

1. Passive Scanning

The client listens silently on each channel without transmitting.

Process:

- a) STA switches to a channel.
- b) Listens for Beacon frames (broadcast periodically by APs, usually every 100 ms).
- c) Collects information from beacons: SSID, BSSID, supported rates, capabilities, security, etc.
- d) Moves to the next channel after a dwell time (MinChannelTime / MaxChannelTime).

Advantages: Saves power (no transmission), less airtime usage.

Disadvantages: Slower (must wait for beacon interval), may miss hidden SSIDs or APs not beacons frequently.

Use Case: Battery-constrained devices preferring power saving.

2. Active Scanning (more common)

Definition: The client actively probes for networks by sending frames.

Process:

- a) STA switches to a channel.
- b) Sends Probe Request frame (broadcast or directed with specific SSID; null SSID for wildcard).
- c) Starts a ProbeTimer and waits for Probe Response frames from APs.
- d) Collects responses containing network details (similar to beacons).
- e) Repeats on all channels.

Advantages: Faster discovery, works well for hidden SSIDs (APs respond to probes), more reliable in noisy environments.

Disadvantages: Consumes more power and airtime due to transmissions.

Probe Request Types: Broadcast (wildcard SSID) or Directed (specific SSID).

After scanning, the STA builds a list of candidate APs and proceeds to authentication/association.

5. Brief about the client association process

The client association process joins a station (STA) to an Access Point (AP) after discovery. It has three main phases:

- a) Scanning— Discover APs.
- b) Authentication — Verify compatibility
 - Usually Open System Authentication (2 frames: Auth Request + Auth Response).
 - Legacy Shared Key (WEP) is rarely used.
- c) Association — Register with the AP.
 - STA sends Association Request (includes capabilities, supported rates, RSN for security, etc.).
 - AP responds with Association Response (status code: success/failure + Association ID — AID).
 - On success, the STA is now "associated" and can exchange data frames.

Reassociation is used during roaming, occurs similar to Association but with user info.

After association, 4-way EAPOL handshake (for WPA2/WPA3) establishes encryption keys. The process ensures the STA and AP agree on capabilities before data transfer begins.

6. Explain each step involved in EAPOL 4-way handshake and the purpose of each keys derived from the process

The EAPOL 4-Way Handshake occurs after association to derive temporal (session) encryption keys for WPA2/WPA3. It uses the Pairwise Master Key (PMK) — derived from PSK or 802.1X/EAP.

Steps:

1. Message 1 (AP - STA): AP sends ANonce (AP nonce) + other info. No key installed yet.
2. Message 2 (STA - AP): STA sends SNonce (STA nonce), MIC (Message Integrity Code) for verification. STA computes PTK.
3. Message 3 (AP - STA): AP sends GTK (Group Temporal Key for multicast) + MIC. Installs PTK.
4. Message 4 (STA - AP): STA confirms installation with MIC. Installs GTK.

Keys Derived:

- PTK (Pairwise Transient Key): Unicast key between AP and one STA. Split into:
 - KCK (Key Confirmation Key): Verifies handshake messages (MIC).
 - KEK (Key Encryption Key): Encrypts GTK delivery.
 - TK (Temporal Key): Actual data encryption key (used with AES-CCMP).
- GTK (Group Temporal Key): Broadcast/multicast key (shared with all STAs in the BSS). Refreshed periodically.

Purpose of Keys:

1. Mutual authentication of keys
2. Fresh key generation (using nonces to prevent replay)
3. Secure installation of encryption keys without exposing the PMK

7. Describe the power saving scheme in MAC layer and explore on the types of Power saving mechanisms

Power Saving in 802.11 MAC allows stations (especially battery-powered clients) to sleep and conserve energy while the AP buffers traffic.

- STA indicates power-save mode in the Power Management bit of Frame Control.
- AP buffers frames for sleeping STAs and announces them in TIM (Traffic Indication Map) inside Beacon frames.
- STA wakes periodically to listen to beacons. If its AID is set in TIM, it polls for data.

Types of Power Saving Mechanisms:

- a) Legacy Power Save (PS-Poll):
 - STA wakes, sends PS-Poll frame to AP.
 - AP sends one buffered frame + More Data bit.
 - STA stays awake until More Data = 0 or sleeps again.
 - Suitable for WSN, Laptops and phones
- b) Automatic Power Save Delivery (APSD) — Enhanced for QoS (802.11e):
 - U-APSD (Unscheduled APSD): STA triggers delivery by sending any uplink frame (e.g., QoS Null). AP delivers all buffered frames for that AC.
 - S-APSD (Scheduled APSD): Delivery at scheduled service periods.
 - Useful for Realtime application and time based applications
- c) TIM-based / Beacon-based:
 - STA wakes at DTIM (Delivery TIM) intervals for broadcast/multicast.
 - Efficient for light traffic.
 - WMM Power Save / U-APSD (most common today): Combines with QoS for better performance.
 - Useful for Laptops, phones (Infrastructure)

8. Describe the Medium Access Control methodologies

802.11 MAC uses contention-based and (legacy) contention-free methods:

a) Distributed Coordination Function (DCF):

- CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance): Stations sense the medium; if idle, they transmit. If busy, they wait using a random backoff timer to avoid collisions.
- Binary Exponential Backoff: To prevent re-collision, stations pick a random wait time from a contention window that doubles with each collision.
- RTS/CTS (Request to Send/Clear to Send): Optional handshake to solve the "hidden node" problem, reserving the channel before transmitting data.
- ACK (Acknowledgment): The receiver sends an ACK frame to confirm successful receipt; if no ACK is received, the transmitter assumes a collision occurred.
- NAV (Network Allocation Vector): A virtual carrier-sensing mechanism where stations maintain a timer indicating when the channel will become free, allowing them to remain quiet during other transmissions.

b) Point Coordination Function (PCF):

- A centralized, polling-based mechanism controlled by an Access Point (AP).
- The AP polls stations to provide contention-free access, typically used for time-sensitive, isochronous traffic.
- PCF operates during a contention-free period, alternating with the contention-based DCF period.

c) Interframe Spaces (IFS):

- SIFS (Short IFS): Highest priority, used for ACK, CTS, or polling responses.
- DIFS (DCF IFS): Used by stations to wait before sensing the channel for contention-based access.

d) Hybrid Coordination Function (HCF) (802.11e):

Introduced in 802.11e to enhance QoS, combining aspects of DCF and PCF to provide prioritized access (EDCA - Enhanced Distributed Channel Access) and parameterized access (HCCA - HCF Controlled Channel Access).

9. Brief about the Block ACK mechanism and its advantages

Block ACK (BA) allows a receiver to acknowledge multiple MPDUs (or an A-MPDU) with a single ACK frame instead of individual ACKs.

- a) Sender requests Block ACK agreement (ADDBA Request/Response).
- b) Multiple frames are sent (often as A-MPDU).
- c) Receiver sends BlockAck frame with a bitmap indicating which MPDUs were received successfully.
- d) Sender retransmits only missing frames (selective repeat).

Types: Immediate Block ACK, Delayed Block ACK.

Advantages:

- Reduces ACK overhead significantly (especially with aggregation).
- Improves throughput in high-data-rate scenarios (802.11n/ac/ax).
- Better efficiency over error-prone wireless links.
- Supports burst transmissions within a TXOP.

Without Block ACK, every frame needs its own ACK, wasting airtime.

10. Explain about A-MSDU, A-MPDU and A-MSDU in A-MPDU

These are aggregation techniques (introduced in 802.11n and enhanced later) to improve efficiency by reducing overhead.

- a) A-MSDU (Aggregated MAC Service Data Unit):
 - Multiple higher-layer MSDUs (e.g., IP packets) are combined into one MPDU.
 - Each subframe has its own header (14 bytes: DA, SA, Length).
 - Aggregated at the upper MAC.
 - Advantage: Reduces per-packet header overhead for small packets. Limited by MPDU size (~2304 bytes typically).
- b) A-MPDU (Aggregated MAC Protocol Data Unit):
 - Multiple complete MPDUs are combined into one PHY transmission.
 - Each MPDU has its own MAC header + FCS, separated by delimiters.
 - Aggregated at the lower MAC.
 - Supports Block ACK for the whole aggregate.
 - Advantage: Reduces PHY preamble/overhead and IFS gaps. Much higher throughput gain.
- c) A-MSDU within A-MPDU:
 - Combines both: Multiple A-MSDUs are packed into individual MPDUs, and then those MPDUs are aggregated into an A-MPDU.
 - Provides maximum efficiency for mixed traffic (small + large packets).
 - Common in modern high-speed WiFi (802.11ac/ax).